Project proposal and submission details

Stroke Risk Factor Analysis Using Patient Health Records

Student / course	Details
Name	Salahuddin Shaikh Prateek Rai Nisarg Sha
Roll number	16014025056 16014025054 16014025055
Branch / semester	EXCP / S.Y. B.Tech / Semester III
Course	Python Programming Laboratory (316U50L301)
Academic year	2026-27

Problem statement — copy into the form

Analyze the statistical associations between recorded stroke status and age, average glucose level, BMI, hypertension and smoking status using a public patient health dataset. Clean and validate the data, summarize the variables, visualize group differences, and apply suitable statistical tests with effect sizes and uncertainty. Assess how missing values and age composition affect the findings, without claiming causation or predicting an individual patient's future stroke risk.

Dataset — upload data/stroke.csv

Stroke Prediction Dataset: 5,110 rows, 12 columns. Required factors: age, average glucose, BMI, hypertension and smoking status. Outcome: stroke (0/1). The supplied raw CSV is below 10 MB.

https://www.kaggle.com/datasets/fedesoriano/stroke-prediction-dataset

Planned methods

Data audit, adult primary analysis, descriptive summaries, eight charts, Welch t-tests, chi-square tests, Holm correction, effect sizes, Wilson intervals and sensitivity checks. Tools: Python, NumPy, Pandas, Matplotlib, Seaborn and SciPy.

Form fields and submission checklist

Form field	What to enter
Email	Use your institutional email account: salahuddin29@somaiya.edu.
Branch	EXCP
Batch	Select your actual assigned batch. The screenshot shows 3; verify it.
Group Number	Use the number assigned by your faculty. Do not infer it from the screenshot.
Problem Statement / Problem Definition	Copy the paragraph on page 1.
Dataset File	Upload data/stroke.csv from this package.
Dataset Link	https://www.kaggle.com/datasets/fedesoriano/stroke-prediction-dataset

Before submission

Check the student name and roll number in project_info.json. Add the confirmed group number, group members and faculty guide if required, then rebuild the PDFs. Review the analysis and run the code so you can explain every result. Follow your faculty's rules for attribution and assistance. If an earlier topic was submitted, use the available edit-response process or contact your faculty to register this topic change.

Which file goes where?

The visible form asks for a dataset, so upload the CSV there. The ZIP is the full capstone package for the final project submission channel; it is not the dataset file. The report and notebook are also supplied separately for convenient review.

Source and limitations

The canonical link above names the dataset. The included CSV was obtained from the public GitHub mirror below. Hospital, population, dates and sampling details are not independently verified. This is an educational statistical analysis; no diagnosis, causal effect or future-risk calculator is claimed.

CSV acquisition mirror on GitHub